



Interpretable physics-informed graph neural networks for flood forecasting

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Abstract

Climate change has intensified extreme weather events, with floods causing significant socioeconomic and environmental damage. Accurate flood forecasting is crucial for disaster preparedness and risk mitigation, yet traditional hydrodynamic models, while precise, are computationally prohibitive for real-time applications. Machine learning surrogates, such as graph neural networks (GNNs), improve efficiency but often lack physical consistency and interpretability. This paper introduces HydroGraphNet, a novel physics-informed GNN framework that, for the first time, integrates the Kolmogorov–Arnold Network (KAN) to enhance model interpretability in unstructured mesh-based flood forecasting. The framework embeds mass conservation laws into the loss function, ensuring physically consistent predictions. Additionally, it employs an autoregressive encoder–processor–decoder architecture that captures spatiotemporal flood dynamics while mitigating error accumulation over long forecasting horizons. Validation on flood data from the White River near Muncie, Indiana, demonstrates a 67% reduction in prediction error, near-zero mass balance error, and a 58% improvement in the critical success index for major flood events compared to a baseline GNN model. These results highlight the potential of the proposed framework to advance real-time flood forecasting with improved physical consistency and interpretability.

1 | INTRODUCTION

Climate change has led to increasingly frequent and severe hydrological extremes, presenting escalating challenges for communities and governments worldwide. Among these disasters, floods—both pluvial, resulting from intense rainfall, and fluvial, arising from river overflow—stand out due to their destructive social, economic, and

environmental impacts (Taheri et al., 2025; Zandsalimi et al., 2024). Reliable and timely predictive capabilities are essential for enhancing disaster preparedness, effective risk mitigation, and supporting timely and accurate actions in anticipation of floods. Additionally, efficient flood models that can be seamlessly integrated into Monte Carlo frameworks to simulate multiple scenarios, enable probabilistic analysis and robust climate adaptation

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strategies. These advancements ultimately safeguard lives, minimize property damage, and guide infrastructure development.

Traditional high-fidelity two-dimensional hydrodynamic models, which solve the shallow water equations (SWE), remain a cornerstone for understanding flood behavior over both space and time. While these models offer valuable insights for risk assessments, their high computational cost restricts their real-time applicability and limits their use in large-scale probabilistic analyses, such as uncertainty quantification or probabilistic flood risk assessment (Caviedes-Voullième et al., 2023; Costabile et al., 2017). Despite advances in high-performance computing and parallelization (Caviedes-Voullième et al., 2023; Hu et al., 2022), numerical constraints and substantial resource demands hinder the scalability of such physics-based approaches for extensive flood scenarios encompassing both pluvial and fluvial events.

This computational bottleneck has driven research toward surrogate modeling techniques, including physics-informed neural networks (PINNs) and convolutional neural networks (CNNs). PINNs incorporate the governing partial differential equations (PDEs) directly into their loss functions, ensuring that their outputs remain physically consistent (Raissi et al., 2019). They have proven valuable in fields such as fluid dynamics (Cai et al., 2021), solid mechanics (Kaewnuratchadasorn et al., 2024), structure analysis and design (Fu et al., 2024; Wang et al., 2024), and environmental modeling (Daw et al., 2022). However, PINNs often rely on computationally expensive automatic differentiation, which hinders their efficiency in complex, high-dimensional applications. CNN-based surrogates have also shown promise in flood mapping (Guo et al., 2021; Kabir et al., 2020), but their reliance on structured grids makes them less suitable for irregular floodplain geometries, which limits their realism and adaptability, particularly in diverse flood scenarios involving both pluvial and fluvial origins.

Other machine learning approaches, such as multilayer perceptrons (Berkhahn et al., 2019) and random forests (Woznicki et al., 2019), have demonstrated skill in flood forecasting; however, they require a fixed input geometry and need retraining when the geometry of the domain changes, potentially reducing their generalizability (Guo et al., 2022; Löwe et al., 2021). Recurrent neural networks, particularly long short-term memory networks, have successfully captured temporal flood dynamics (Kao et al., 2021; Nevo et al., 2022), yet they struggle with unstructured spatial data and lack an inherent framework to ensure physical consistency.

Graph neural networks (GNNs) present a compelling alternative, as they naturally handle unstructured meshes by representing floodplains as graphs whose nodes corre-

spond to computational points and whose edges encode spatial relationships. Through localized message passing, GNNs capture complex spatiotemporal dependencies and have shown great promise in modeling fluid dynamics (Bentivoglio et al., 2023; Pfaff et al., 2020; Sanchez-Gonzalez et al., 2020). For instance, SWE-GNN integrates GNNs with finite-volume methods to simulate flood evolution accurately and with physical transparency (Bentivoglio et al., 2023). Nonetheless, conventional GNNs remain data-driven, sometimes leading to physically implausible solutions that violate conservation laws. Moreover, error accumulation during autoregressive predictions and limited interpretability pose additional challenges.

In response, this study introduces HydroGraphNet, the first physics-informed GNN framework tailored for both pluvial and fluvial flood forecasting. By merging GNNs' inherent suitability for unstructured domains with a physics-informed training objective, the predictions are ensured to obey conservation laws such as mass balance without relying on expensive automatic differentiation. Instead, physical principles are incorporated directly into the loss function through discrete comparisons, which significantly enhances computational efficiency and physical realism.

Furthermore, the proposed approach integrates the pushforward trick (Brandstetter et al., 2022) for the first time to reduce error propagation over long prediction horizons and improve the stability and reliability of autoregressive forecasting. Also, the Kolmogorov–Arnold network (KAN) framework is adopted (Z. Liu et al., 2025) for the first time to increase model interpretability, elucidating how environmental variables, such as elevation, slope, and Manning's roughness coefficient, influence flood dynamics across different flood types. More importantly, KAN implicitly contributes to physically plausible predictions by capturing hydrologically meaningful relationships between terrain features and water flow behavior, which reduces the likelihood of physically inconsistent outputs.

The HydroGraphNet framework is validated using a real-world case study of flooding in the White River near Muncie, Indiana, encompassing both pluvial and fluvial events. Compared to a baseline GNN, the HydroGraphNet model achieves markedly higher predictive accuracy, identifies major flood events more reliably, and maintains near-zero mass balance errors—an indicator of robust physical consistency. It also accurately captures flood arrival times, closely matching results from high-fidelity hydrodynamic simulations. Together, these findings highlight the model's potential for delivering efficient, reliable, and interpretable flood forecasts suited to operational environments dealing with diverse flood scenarios.

The primary contributions of this paper are as follows:



- The first physics-informed GNN framework is introduced for flood forecasting, addressing both pluvial and fluvial events and ensuring physical consistency without the need for expensive automatic differentiation.
- The pushforward trick is implemented to minimize error propagation in long-term autoregressive predictions, enhancing forecast stability and reliability.
- The KAN framework is adopted to improve model interpretability and implicitly enhance physical plausibility by learning structured, hydrologically meaningful transformations of environmental variables.
- The effectiveness of the proposed approach is demonstrated through a real-world case study that integrates multiple complexities:
 - The case study captures both pluvial and fluvial flooding dynamics, demonstrating the model's capability to handle multiple flood types within a unified framework.
 - It focuses on a large-scale domain with an extensive computational mesh, comprising 4787 nodes, ensuring the model's applicability to realistic flood scenarios.
 - The study includes urban areas, accounting for the challenges posed by intricate terrain features, variable land use, and spatial heterogeneity.
 - It leverages diverse hydrograph and hyetograph scenarios, highlighting the model's robustness under varying environmental and boundary conditions.

The remainder of this paper is organized as follows: Section 2 reviews state-of-the-art approaches and relevant literature on GNNs and other deep learning methods in flood forecasting. Section 3 outlines the key theoretical concepts underlying the proposed approach. Section 4 details the proposed HydroGraphNet methodology for flood prediction, and Section 5 demonstrates the experimental results and discusses the model's performance. Finally, Section 6 concludes the paper and proposes avenues for future research.

2 | RELATED WORK

Flood modeling has witnessed significant advancements through the adoption of deep learning techniques, progressing from simpler models like multi-layer perceptron (MLPs) to advanced frameworks such as GNNs. This section critically reviews these approaches, focusing on their strengths, limitations, and contributions to addressing challenges in flood modeling.

MLPs were among the earliest machine learning models used in flood mapping and susceptibility analysis. They were applied to tasks such as superresolution mapping

of flood extents and flood susceptibility mapping based on conditioning factors like slope, elevation, and rainfall (Chakraborty et al., 2022; Khoirunisa et al., 2021; Li, Chen, et al., 2016; Li et al., 2015; Li, Xu, et al., 2016). While effective for static and small-scale datasets, MLPs are fundamentally limited by their inability to capture spatial relationships within floodplains. Their reliance on structured, preprocessed data inputs reduces adaptability and scalability to dynamic scenarios or irregular floodplain geometries. Consequently, MLPs have largely been replaced by models better suited to exploit spatial and temporal dependencies inherent in flood dynamics.

CNNs advanced flood mapping by leveraging their ability to extract spatial features from structured data, such as satellite or UAV imagery (Zandsalimi et al., 2025). Studies such as Nogueira et al. (2018) and Kang et al. (2018) employed CNNs for flood classification, while encoder-decoder architectures enabled more detailed flood extent mapping (Hashemi-Beni & Gebrehiwot, 2021; Wieland & Martinis, 2019).

Despite their effectiveness, CNNs faced critical limitations in representing irregular floodplain geometries. Their reliance on structured grids constrained their adaptability to real-world scenarios where floodplains often have complex, unstructured terrains. Furthermore, most CNN-based models primarily focused on static flood mapping rather than dynamic forecasting. Attempts to integrate temporal dependencies, such as combining CNNs with gated recurrent units (GRU) for flood forecasting (Dong et al., 2021), showed promise but suffered from error accumulation over long prediction horizons, highlighting the need for more robust architectures.

PINNs have emerged as a promising approach by embedding the governing equations directly into the training process, thereby bridging traditional hydrodynamic modeling and machine learning. PINNs have been successfully applied to a range of hydrodynamic tasks—including solving the Euler equations for aerodynamic flows (Mao et al., 2020), modeling fluid flow in porous media (Almajid & Abu-Al-Saud, 2022), and predicting water levels or river flows (Donnelly et al., 2024; B. Liu et al., 2022). However, despite their demonstrated ability to maintain physical consistency, PINNs typically rely on automatic differentiation to compute PDE residuals, which can be computationally expensive and memory intensive for large-scale, high-dimensional applications. Furthermore, although GPU acceleration can alleviate some of these costs in small- or medium-scale problems, PINNs often require extensive retraining when there are changes in domain geometry or boundary conditions, such as alterations in inflow hydrographs or rainfall



TABLE 1 Comparison of state-of-the-art methods and HydroGraphNet.

Method	Unstructured meshes	Physical consistency	Interpretability
MLP	✗	✗	✗
CNN	✗	✗	✗
PINN	✗	✓	✗
GNN	✓	✗	✗
HydroGraphNet	✓	✓	✓

Abbreviations: CNN, convolutional neural network; GNN, graph neural network; MLP, multi-layer perceptron; PINN, physics-informed neural network.

patterns. Additionally, the structured nature of PINNs limited their application to irregular floodplain geometries, which makes them less adaptable to the complexities of flood modeling.

GNNs have recently emerged as a powerful tool for flood modeling due to their ability to handle unstructured data and capture spatial dependencies. GNNs represent floodplains as graphs, where nodes correspond to computational points and edges encode spatial relationships. This flexibility makes them particularly suitable for modeling complex flood dynamics in irregular terrains.

Bentivoglio et al. (2023) and its extension in Bentivoglio et al. (2024) highlighted the potential of hydraulic-based GNNs for simulating flood propagation, particularly in dam-break scenarios. These models incorporated finite-volume methods and introduced innovations such as multiscale processing, time-varying boundary conditions, and rotational invariance, enhancing their generalization and computational stability. Despite these advancements, the models lacked explicit physical constraints, such as mass conservation. The models were also limited in scope and focused narrowly on dam-break flooding. This restricts their applicability to more complex floods where diverse interactions between flow and terrain occur.

More recently, Kazadi et al. (2024) proposed FloodGNN-GRU, a spatiotemporal model that combines GNNs with GRU to capture temporal dynamics. The model was trained and tested using simulation data from Hurricane Harvey in Houston. While the model achieved high computational efficiency and predictive accuracy compared to traditional machine learning approaches, it was developed and validated based on a single simulation event. This reliance on data from a fixed hydrograph raises questions about its generalizability to other flood scenarios, as it has not been evaluated on events with different rainfall or boundary conditions. Moreover, FloodGNN-GRU does not include explicit physics-based constraints, leading to

predictions that may violate fundamental principles, such as mass conservation.

In addition to the fact that most existing deep learning-based flood prediction approaches struggle to enforce fundamental physical principles, which can lead to physically implausible predictions, the majority of these models are trained and evaluated for short-term forecasting. As a result, they are prone to instability when rolled out over longer prediction horizons (Liao et al., 2023; Roy et al., 2025). Furthermore, despite their ability to capture complex spatiotemporal dependencies, these models function as black boxes, lacking interpretability and failing to provide insights into the key environmental factors driving flood dynamics.

To address these gaps, we introduce HydroGraphNet to produce physically consistent, stable, and interpretable flood forecasts. Specifically, HydroGraphNet improves flood predictions by (1) embedding physical constraints in the loss function, (2) leveraging the pushforward trick to improve forecast stability for long-term predictions, and (3) integrating the KAN to enhance interpretability. Table 1 provides a brief overview of the advantages offered by HydroGraphNet compared to other available methods in the literature.

3 | BACKGROUND

This section outlines the mathematical foundations supporting the proposed model. First, numerical modeling approaches for flood simulation are discussed, focusing on the shallow water equations and their numerical solution via the finite-volume method. Next, the fundamentals of GNNs are presented. Subsequently, the graph network (GN) block is described as a pivotal component of the framework, followed by the encoder–processor–decoder architecture. Finally, the KAN is introduced along with its role in enhancing the interpretability and efficiency of the proposed model.

3.1 | Numerical modeling

3.1.1 | Shallow water equations

Flood dynamics can be described by the SWE, which forms a system of hyperbolic partial differential equations enforcing mass and momentum conservation (Vreugdenhil, 2013). Assuming negligible vertical accelerations, the two-dimensional SWE can be written as

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot \mathbf{F} = \mathbf{s}, \quad (1)$$



where $\mathbf{u}$ is the vector of conserved variables, $\mathbf{F}$ is the flux matrix, and $\mathbf{s}$ is the source term vector. These are given by

$$\mathbf{u} = \begin{bmatrix} h \\ q_x \\ q_y \end{bmatrix}, \mathbf{F} = \begin{bmatrix} q_x & q_y \\ \frac{q_x^2}{h} + \frac{gh^2}{2} & \frac{q_x q_y}{h} \\ \frac{q_x q_y}{h} & \frac{q_y^2}{h} + \frac{gh^2}{2} \end{bmatrix}, \mathbf{s} = \begin{bmatrix} 0 \\ gh(s_{0x} - s_{fx}) \\ gh(s_{0y} - s_{fy}) \end{bmatrix}, \quad (2)$$

where h is the water depth and $q_x = uh$, $q_y = vh$ are the depth-integrated discharge components in the x - and y -directions, respectively, where u and v are the depth-averaged velocities. The scalar g denotes the gravitational acceleration. The source terms s_{0x} , s_{0y} represent the bed slope components, and s_{fx} , s_{fy} represent frictional losses in the respective directions.

3.1.2 | Finite-volume method

Analytical solutions to the SWE are generally not feasible unless simplified extensively. Instead, numerical methods such as the finite-volume method are commonly employed (Alcrudo & Garcia-Navarro, 1993). In this approach, the spatial domain is discretized into finite cells. Within each cell, the hydraulic variables (h , q_x , q_y) are approximated at the cell center. Integrating the governing equations over a cell and applying the divergence theorem yields a discrete-time update equation:

$$\mathbf{u}_i^{t+1} = \mathbf{u}_i^t + \Delta t \left(\mathbf{s}_i - \sum_{j=1}^{N_i} (\mathbf{F} \cdot \mathbf{n})_{ij} \frac{l_{ij}}{a_i} \right), \quad (3)$$

where $\mathbf{u}_i^t$ represents the variables in cell i at time t , a_i is the cell area, N_i is the number of neighboring cells, l_{ij} is the length of the shared interface with the j th neighbor, and $(\mathbf{F} \cdot \mathbf{n})_{ij}$ represents the flux through that interface, with $\mathbf{n}$ being the outward unit normal vector.

To ensure stability in explicit schemes, the Courant–Friedrichs–Lewy (CFL) condition must be satisfied (Courant et al., 1967):

$$C = \frac{v \Delta t}{\Delta x} \leq 1, \quad (4)$$

where v is the characteristic wave speed, Δt is the time step, and Δx is the spatial resolution. While small time steps ensure stability, they also increase the computational cost, thus motivating the exploration of more efficient numerical methods.

Interestingly, the finite-volume formulation offers a natural stepping stone toward more flexible, data-driven approaches. Each computational cell and its relationships with neighboring cells can be viewed as forming a network

of interconnected elements. In this network, fluxes transmitted across cell interfaces resemble localized exchanges of information between elements. Reinterpreting the computational domain in this manner provides a conceptual bridge to graph-structured representations. Here, nodes (cells) exchange messages (fluxes) along edges (interfaces), a perspective that aligns closely with the mechanisms of graph-based methods. These methods, as described in subsequent sections, leverage such relational structures to efficiently learn complex spatiotemporal dynamics directly from data.

3.2 | Graph neural networks

GNNs provide a powerful framework for addressing the computational challenges of traditional methods, particularly for flood modeling on irregular terrains. Unlike CNNs, which process regular, grid-structured data like images, GNNs are designed for graph-structured data where nodes and edges form non-Euclidean structures (Khayambashi et al., 2024; Taghizadeh, Khayambashi, et al., 2024). This capability allows GNNs to naturally handle the irregular meshes commonly used in flood simulations.

The central operation in GNNs is the message-passing mechanism, which iteratively updates node and edge features by aggregating information from neighboring nodes (Doctorarastoo et al., 2024). Several GNN architectures have been developed, including graph convolutional networks (Kipf & Welling, 2016), GraphSAGE (Hamilton et al., 2017), and graph attention networks (Veličković et al., 2017), each differing in their approach to message aggregation. GNNs' flexibility makes them particularly suitable for capturing local and global dependencies in spatiotemporal flood dynamics.

3.3 | Graph network block

A key component of the proposed framework is the GN block, which operates on a graph $G = (V, E)$, where V and E denote the sets of nodes and edges, respectively. Each node $v \in V$ is associated with a feature vector $\mathbf{v}$, and each edge $e \in E$ is linked with a feature vector $\mathbf{e}$. The GN block updates these features through a three-step process. First, the edge features are updated based on the attributes of the connected nodes:

$$\mathbf{e}'_{ij} = \psi_e(\mathbf{e}_{ij} \oplus \mathbf{v}_i \oplus \mathbf{v}_j), \quad (5)$$

where ψ_e is an MLP and $\oplus$ denotes concatenation. Next, the updated edge features are aggregated at each node to



form a message:

$$\mathbf{m}_i = \sum_{j \in \mathcal{N}(i)} \mathbf{e}'_{ij}, \quad (6)$$

where $\mathcal{N}(i)$ is the set of neighboring nodes for node i . Finally, the node features are updated using the aggregated message and the original node attributes:

$$\mathbf{v}'_i = \psi_v(\mathbf{v}_i \oplus \mathbf{m}_i), \quad (7)$$

where ψ_v is another MLP. This iterative process embeds relational information into nodes and edges, progressively expanding the effective neighborhood for each node.

3.4 | Encoder–processor–decoder model

The encoder–processor–decoder model utilizes the flexibility of GNNs to process graph-structured data effectively (Pfaff et al., 2020; Sanchez-Gonzalez et al., 2020). As illustrated in Figure 1a, the architecture consists of three key components: the encoder, which initializes node and edge features; the processor, which iteratively refines these features; and the decoder, which maps the refined features to the desired output.

The *encoder* processes the input node and edge features to create an initial hidden representation. For nodes, the encoder integrates input features $\mathbf{u}$, global features ξ , and boundary indicators $\mathbf{b}_G$ as follows:

$$\mathcal{E}_v(\mathbf{u}, \xi, G) := \psi_{\xi}^v(\mathbf{u} \oplus \xi \oplus \mathbf{b}_G), \quad (8)$$

where $\oplus$ denotes concatenation and ψ_{ξ}^v is an MLP mapping these inputs into a hidden-state space. For edges, the encoder computes features based on relative positions between connected nodes:

$$\mathcal{E}_e(G) := \psi_{\xi}^e(\mathbf{e}_G), \quad (9)$$

where $\mathbf{e}_G$ represents edge attributes derived from node coordinates, including relative distances and edge lengths.

The *processor* comprises multiple message-passing layers that iteratively refine node and edge features. Each layer aggregates information from neighboring nodes and edges, enabling the model to capture complex dependencies across the graph. Letting $\mathbf{h}_1^v$ and $\mathbf{h}_1^e$ denote the initial node and edge features from the encoder, the processor updates them iteratively:

$$\mathcal{P}(\mathbf{v}, \mathbf{e}, G) := F_j(\mathbf{h}_j^v, \mathbf{h}_j^e, G), \quad (10)$$

where F_j represents the computations in the j th layer, progressively refining $\mathbf{h}_j^v$ and $\mathbf{h}_j^e$ through m message passing layers.

Finally, the *decoder* translates the processed node features into the target output space. Using the refined node features $\mathbf{v}'$, the decoder applies an MLP to produce the desired output:

$$\mathcal{D}(\mathbf{v}') := \psi_{\mathcal{D}}(\mathbf{v}'), \quad (11)$$

where $\psi_{\mathcal{D}}$ maps the hidden features back to the output feature space.

The end-to-end computational flow of the encoder–processor–decoder model can be summarized as

$$\phi(\mathbf{u}, \xi, G) := \mathcal{D}(\mathcal{P}(\mathcal{E}_v(\mathbf{u}, \xi, G), \mathcal{E}_e(G), G)), \quad (12)$$

where ϕ represents the overall mapping from input to output. This architecture provides a powerful framework for modeling complex spatiotemporal phenomena, which balances flexibility and computational efficiency.

3.5 | Kolmogorov–Arnold network

KANs are a novel class of neural networks inspired by the Kolmogorov–Arnold representation theorem, which asserts that any multivariate continuous function can be expressed as a finite composition of univariate functions and the binary operation of addition (Z. Liu et al., 2025). This insight forms the foundation of KANs, where instead of using fixed linear weight parameters as in traditional neural networks, the model replaces them with learnable univariate functions, typically implemented as splines. This structural modification allows KANs to flexibly adapt to complex patterns in data while preserving a clear mathematical structure that enhances interpretability.

Given a supervised learning task with input–output pairs $\{x_i, y_i\}$, a KAN models the target function $f(x)$ as

$$f(x) = \sum_{q=1}^{2n+1} \Phi_q \left(\sum_{p=1}^n \phi_{q,p}(x_p) \right), \quad (13)$$

where $\phi_{q,p}$ and Φ_q are trainable univariate functions parameterized as splines. Unlike traditional MLPs, which consist of layers of neurons where information flows through a series of linear transformations followed by fixed nonlinear activation functions, KANs treat these

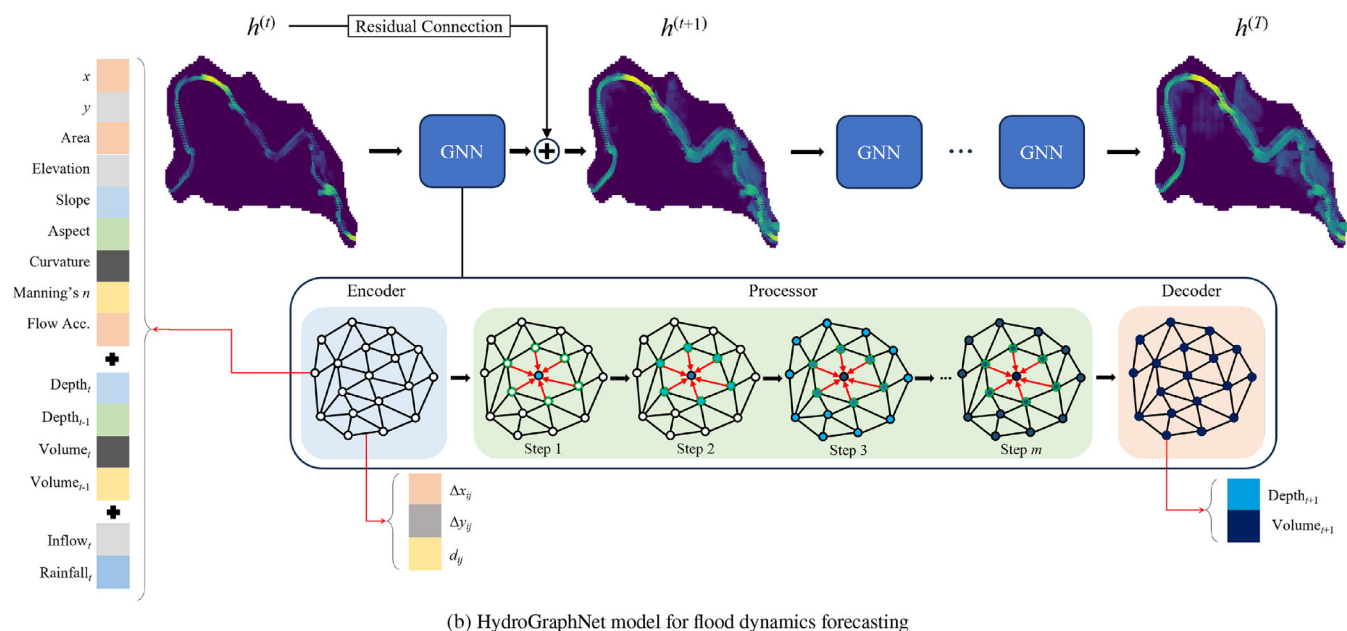
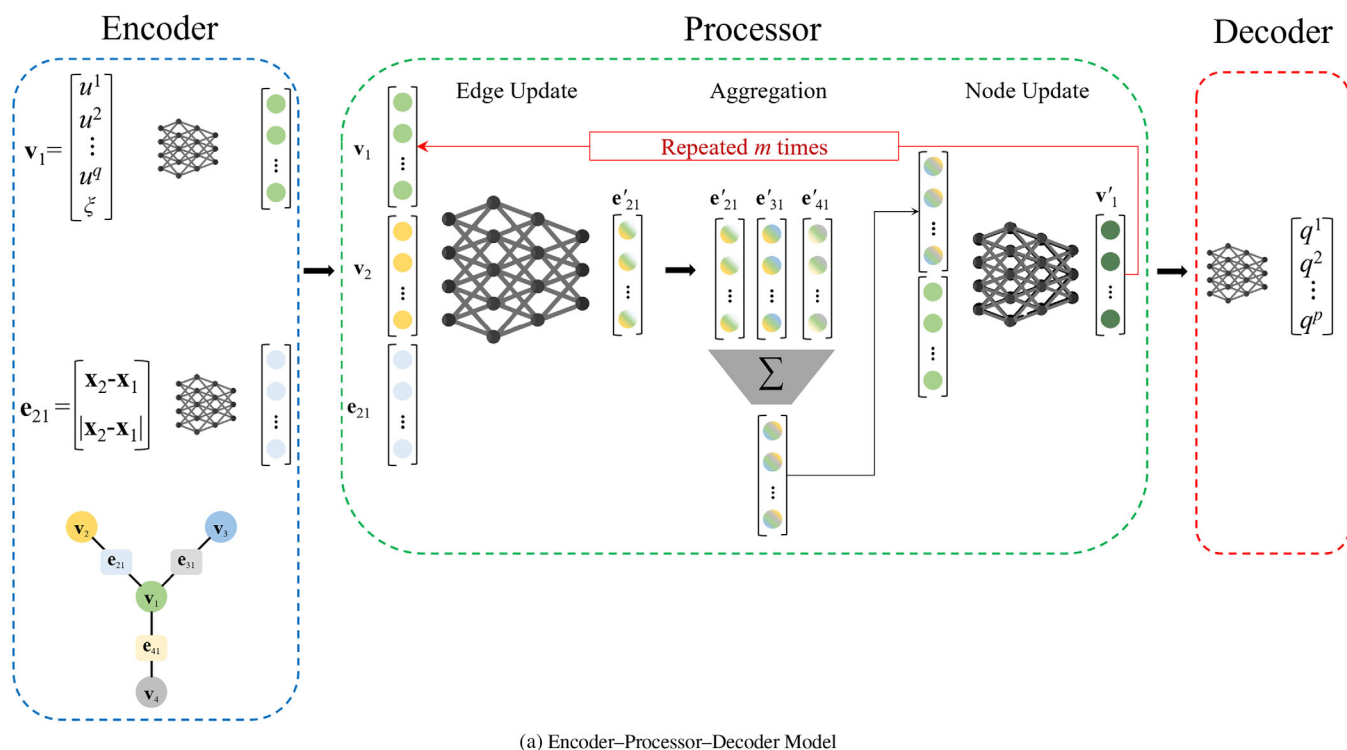


FIGURE 1 (a) Schematic of the encoder–processor–decoder model. The architecture consists of three main components: the encoder, which initializes the node and edge features from the input data; the processor, which refines these features iteratively through message-passing steps; and the decoder, which maps the refined features to the target outputs. Note that $\mathbf{x}$ denotes the vector of spatial coordinates (x, y). (b) The autoregressive encoder–processor–decoder architecture of the HydroGraphNet model is used for flood dynamics forecasting. Each GNN block refines spatiotemporal representations of nodes and predicts the next water depth and volume, utilizing message-passing and residual connections.



components holistically. Instead of static weights, KANs utilize function matrices that dynamically adjust to input data, making them more adaptable and expressive. The transformation at each layer in a KAN follows:

$$x_{l+1,j} = \sum_{i=1}^{n_l} \phi_{l,j,i}(x_{l,i}), \quad (14)$$

where $\phi_{l,j,i}$ represents the spline-based activation function that maps node i in layer l to node j in layer $l + 1$. By replacing static weight parameters with functions, KANs eliminate the need for fixed nonlinearities like a rectified linear unit (ReLU) or sigmoid, allowing each transformation to be data-dependent and continuously adaptable.

This fundamental difference between KANs and MLPs leads to significant advantages. In MLPs, the relationship between inputs and outputs is often difficult to interpret because the transformation at each layer is a combination of matrix multiplications and predefined activation functions, which makes it challenging to extract explicit functional relationships from the model. In contrast, KANs retain a structure where each transformation remains an interpretable function of the input variables. The learned splines can be visualized to understand how the model processes different inputs, providing insights into the functional dependencies captured by the network. This transparency makes KANs particularly appealing for scientific and engineering applications, where interpretability is essential for trust and validation.

Beyond interpretability, KANs offer superior modeling capability, particularly for capturing compositional structures in data. Many real-world phenomena, such as hydrological and meteorological processes, exhibit hierarchical dependencies that are challenging for traditional MLPs to capture efficiently. By leveraging the Kolmogorov–Arnold framework, KANs naturally encode these dependencies, allowing them to model complex spatiotemporal patterns with greater precision. This makes them well-suited for applications like flood forecasting, where understanding the interactions between hydrological variables is crucial. By incorporating KANs into the proposed framework, the ability to learn high-fidelity, physically consistent representations of flood dynamics is enhanced, offering a more robust and interpretable alternative to conventional deep learning models.

4 | METHODOLOGY

The proposed framework employs an autoregressive encoder–processor–decoder architecture to predict water

depth and volume across multiple future time steps, capturing the temporal evolution of both fluvial and pluvial flood dynamics while embedding physical constraints within the GNN model. This section covers the graph representation, the autoregressive architecture, the physics-informed loss function, the pushforward trick for model stability, and the integration of the KAN framework for interpretability.

4.1 | Graph representation

In this framework, the floodplain is represented as a graph $G = (V, E)$, where V denotes the set of nodes and E denotes the edges connecting neighboring nodes. Each node $v \in V$ represents a mesh point in the floodplain, and it is associated with a feature vector $\mathbf{x}_v \in \mathbb{R}^d$, comprising both static and dynamic attributes. The global feature vector $\xi \in \mathbb{R}^s$, representing the external forcing conditions, includes the flow hydrograph at the upstream boundary (for fluvial floods) and rainfall intensity over time (for pluvial floods). This vector is shared across all nodes at each time step. The node features $\mathbf{x}_v$ at each node v include

- *Static node features*: These are invariant over time and consist of spatial coordinates (x, y) , catchment area, elevation, slope, aspect, curvature, Manning's roughness coefficient, flow accumulation, and imperviousness (Löwe et al., 2021).
- *Dynamic node features*: These vary over time and include water depth and volume at previous time steps up to the current time t , contributing to a historical view of flood evolution across the domain.

Edge features $\mathbf{e}_{ij} \in \mathbb{R}^{d_e}$ for $e_{ij} \in E$ represent spatial relationships between nodes v_i and v_j :

$$\mathbf{e}_{ij} = [\Delta x_{ij}, \Delta y_{ij}, d_{ij}], \quad (15)$$

where $\Delta x_{ij} = x_i - x_j$, $\Delta y_{ij} = y_i - y_j$, and d_{ij} is the Euclidean distance between nodes v_i and v_j . These features capture relative positions and distances, essential for modeling spatial interactions.

4.2 | Autoregressive encoder–processor–decoder architecture

The GNN framework uses an autoregressive encoder–processor–decoder architecture, illustrated in Figure 1b, to predict water depth and volume across future time steps,



capturing spatiotemporal dependencies while leveraging residual connections for accuracy. The output $\hat{\mathbf{y}}^{(t+1)}$ at time $t + 1$ is computed as

$$\hat{\mathbf{y}}^{(t+1)} = \mathcal{D}\left(\mathcal{P}\left(\mathcal{E}_v\left(\mathbf{x}_v^{(t)}, \xi, G\right), \mathcal{E}_e(G), G\right)\right) + \mathbf{y}^{(t)}, \quad (16)$$

where $\mathcal{E}_v$ encodes node features (both static and dynamic attributes) and $\mathcal{E}_e$ encodes edge features representing spatial relationships. The processor $\mathcal{P}$ applies message-passing layers to aggregate neighborhood information, enhancing the spatiotemporal context of each node. Finally, the decoder $\mathcal{D}$ maps these processed features to the output space and predicts the incremental changes in water depth and volume at each node relative to the previous time step. The predicted changes are then added to the previous time step's values, $\mathbf{y}^{(t)}$, to compute the water depth and volume at time $t + 1$.

4.3 | Physics-informed loss function

To enforce mass conservation in flood forecasting, a physics-informed loss function is designed with several components to maintain physical consistency:

4.3.1 | Prediction loss

This term calculates the mean squared error between the predicted and true water depth and volume. The volume component is normalized by the catchment area A_v at each node to ensure consistency across spatial scales:

$$\mathcal{L}_{\text{pred}} = \frac{1}{N_t} \sum_{t=1}^{N_t} \left(\frac{1}{|V|} \sum_{v \in V} \left[\left(\hat{h}_v^{(t)} - h_v^{(t)} \right)^2 + \left(\frac{\hat{v}_v^{(t)}}{A_v} - \frac{v_v^{(t)}}{A_v} \right)^2 \right] \right), \quad (17)$$

where N_t represents the number of prediction steps, $|V|$ indicates the number of nodes, $\hat{h}_v^{(t)}$ and $h_v^{(t)}$ are the predicted and true water depths, and $\hat{v}_v^{(t)}$ and $v_v^{(t)}$ are the predicted and true volumes at node v .

4.3.2 | Physics-based loss

To enforce mass conservation over the entire domain, a physics-based loss function is introduced, derived from the continuity equation used in hydrodynamic simulations. The continuity equation ensures that the change in storage within a domain is balanced by the net sources and sinks, conceptually expressed as

$$\Delta(\text{storage}) = \Delta(\text{sources} - \text{sinks}). \quad (18)$$

The schematic in Figure 2 illustrates the principle of mass conservation applied in this work. The predicted volumes and inflows between successive time steps must satisfy the continuity equation constraints.

In this context, $s^{(t)}$ represents the net inflow rate at time t . For fluvial flooding scenarios, $s^{(t)}$ corresponds to the flow hydrograph at the upstream boundary condition. For fluvial-pluvial cases, $s^{(t)}$ includes the sum of the flow hydrograph at the upstream boundary condition and the rainfall input over the total domain area:

$$s^{(t)} = i_{\text{upstream}}^{(t)} + r^{(t)} \cdot A_{\text{total}}, \quad (19)$$

where $i_{\text{upstream}}^{(t)}$ is the upstream boundary flow rate, $r^{(t)}$ is the rainfall intensity at time t , and A_{total} is the total area of the domain.

Inspired by the approach of Donnelly et al. (2024), two inequalities are established that enforce different aspects of the continuity equation.

The first inequality ensures that the total predicted volume at time t does not exceed the ground truth volume at the previous time step plus the average inflow between $t - 1$ and t , accounting for the average infiltration rate times the area for each node:

$$\sum_{v \in V} \hat{v}_v^{(t)} \leq \sum_{v \in V} v_v^{(t-1)} + \Delta t \cdot \left(\frac{s^{(t-1)} + s^{(t)}}{2} - \sum_{v \in V} \frac{i_v^{(t-1)} + i_v^{(t)}}{2} \cdot A_v \right), \quad (20)$$

where $i_v^{(t)}$ represents the infiltration rate for node v at time t . This inequality represents a forward consistency check, ensuring that the predicted volume at t accounts for the initial storage (from the ground truth at $t - 1$), the inflow during the interval, and the loss due to infiltration.

The second inequality checks that the ground truth volume at the next time step does not exceed the total predicted volume at time t plus the average inflow between t and $t + 1$, again accounting for the average infiltration rate times the area for each node:

$$\sum_{v \in V} v_v^{(t+1)} \leq \sum_{v \in V} \hat{v}_v^{(t)} + \Delta t \cdot \left(\frac{s^{(t)} + s^{(t+1)}}{2} - \sum_{v \in V} \frac{i_v^{(t)} + i_v^{(t+1)}}{2} \cdot A_v \right). \quad (21)$$

This backward consistency check ensures that the predicted volume at time t aligns with the storage requirements (from the ground truth at $t + 1$), the inflow balance, and the losses due to the average infiltration over the time interval.

To incorporate these inequalities into the loss function and make them differentiable, the ReLU function is used, defined as $\text{ReLU}(x) = \max(0, x)$. The physics-based loss function is formulated as

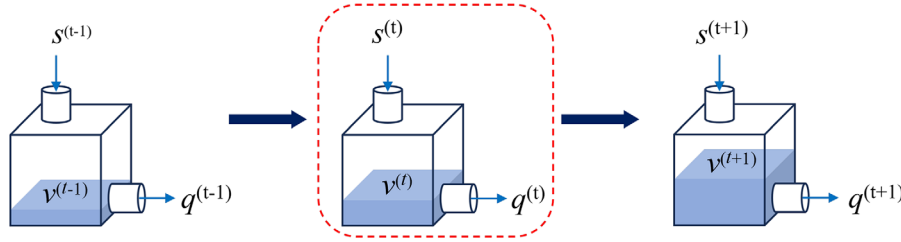


FIGURE 2 Illustration of mass conservation in the context of hydrodynamic simulations, as enforced by Equations (20) and (21).

$$\mathcal{L}_t^{(1)} = \left(\max \left(0, \frac{\sum_{v \in V} \hat{v}_v^{(t)} - \sum_{v \in V} v_v^{(t-1)} - \Delta t \cdot \left(\frac{s^{(t-1)} + s^{(t)}}{2} - \sum_{v \in V} \frac{i_v^{(t-1)} + i_v^{(t)}}{2} \cdot A_v \right)}{A_{\text{total}}} \right) \right)^2 \quad (22)$$

$$\mathcal{L}_t^{(2)} = \left(\max \left(0, \frac{\sum_{v \in V} v_v^{(t+1)} - \sum_{v \in V} \hat{v}_v^{(t)} - \Delta t \cdot \left(\frac{s^{(t)} + s^{(t+1)}}{2} - \sum_{v \in V} \frac{i_v^{(t)} + i_v^{(t+1)}}{2} \cdot A_v \right)}{A_{\text{total}}} \right) \right)^2 \quad (23)$$

$$\mathcal{L}_{\text{physics}} = \frac{1}{N_t} \sum_{t=1}^{N_t} [\mathcal{L}_t^{(1)} + \mathcal{L}_t^{(2)}]. \quad (24)$$

This loss function penalizes any violation of the inequalities in Equations (20) and (21), ensuring that the predicted volumes adhere to the principles of mass conservation. Minimizing $\mathcal{L}_{\text{physics}}$ ensures that the change in total volume between successive time steps is consistent with the net inflow.

The inclusion of ReLU functions allows the loss to focus only on scenarios where the inequalities are violated, effectively ignoring cases where mass conservation is naturally satisfied. Squaring the terms ensures smoothness and differentiability, which is beneficial for optimization algorithms used during training.

It must be noted that while the physics-informed loss function is formulated at the global level to enforce total volume conservation across the floodplain, it inherently influences local flow dynamics through the message-passing mechanism of the GNN. In fact, enforcing strict conservation at the individual node level could introduce numerical instability due to subgrid heterogeneities in terms of terrain, infiltration, and flow accumulation. However, the GNN's relational structure naturally redistributes water across neighboring nodes in a physically consistent manner. Thus, while there is no explicit local flow constraint, the learned redistribution process, guided by

the global mass balance, preserves the physicality of both global and local flood dynamics.

Another essential physical constraint enforced in the loss function is the relationship between predicted water volume and water depth within each computational cell. Since the model simultaneously predicts both quantities, it is crucial to maintain their physical consistency. Specifically, the model predicts the maximum water depth, $\hat{h}_v^{(t)}$, at each cell, which represents the highest water level within that cell. The average water depth is computed as the water volume per unit area, $\frac{\hat{v}_v^{(t)}}{A_v}$. To maintain physical plausibility, the mean water depth must not exceed the maximum depth, as this would imply an unrealistic water distribution. Enforcing this constraint ensures that the model produces hydrologically coherent predictions.

To enforce this physical principle during model training, we define a penalty term that activates only when the inequality $\frac{\hat{v}_v^{(t)}}{A_v} \leq \hat{h}_v^{(t)}$ is violated. This penalty is formulated using a ReLU function to capture only positive discrepancies, and the deviation is squared to ensure that larger violations incur a disproportionately higher penalty while maintaining smooth gradients for stable optimization.



Specifically, the penalty is defined as

$$\mathcal{L}_{\text{penalty}} = \frac{1}{N_t} \sum_{t=1}^{N_t} \left(\frac{1}{|V|} \sum_{v \in V} \left[\max \left(0, \frac{\hat{v}_v^{(t)}}{A_v} - \hat{h}_v^{(t)} \right) \right]^2 \right). \quad (25)$$

By averaging this penalty across all time steps, N_t , and computational cells, $v \in V$, we ensure that the constraint is enforced globally throughout the model's predictions. This penalty term is critical as it not only enforces a fundamental hydrological consistency but also enhances the interpretability of the model by ensuring that the predictions adhere to known physical behavior.

Finally, the one-step prediction loss, $\mathcal{L}_{\text{one-step}}$, is defined as the sum of the prediction loss, physics-based loss, and penalty loss:

$$\mathcal{L}_{\text{one-step}} = \mathcal{L}_{\text{pred}} + \mathcal{L}_{\text{physics}} + \mathcal{L}_{\text{penalty}}. \quad (26)$$

4.4 | Pushforward trick for stability

In autoregressive forecasting, the model makes a prediction at each time step based on its own previous predictions rather than on the ground-truth state. Over multiple time steps, any small prediction errors can accumulate and push the model's inputs into regions of the input space it never saw during training, which is a problem often referred to as *distribution shift*. This shift can cause *instability* in the predictions, meaning errors may grow uncontrollably over longer horizons.

To address this, we adopt the *pushforward trick* (Brandstetter et al., 2022), which exposes the model to perturbed inputs during training so that it learns to be robust against small deviations in its own predictions. In essence, instead of always conditioning on the perfect ground truth at each step, we let the model observe and correct for its own prediction errors.

Mathematically, we define the prediction process as an operator $\mathcal{A}$, which in this case corresponds to the Hydro-GraphNet model, mapping the system state at time t , denoted as $\hat{u}^{(t)}$, to the next time step:

$$\hat{u}^{(t+1)} = \mathcal{A}(\hat{u}^{(t)}). \quad (27)$$

Due to small errors accumulating over time, the actual predicted state distribution deviates from the true distribution $u^{(t+1)}$, leading to instability.

The pushforward trick introduces a stability loss term that forces the model to be robust against small deviations in its own predictions. Instead of training only on the ideal ground truth, the model is exposed to perturbed inputs, simulating the kind of distribution shift that occurs

during real rollouts. This teaches the model to correct for deviations, preventing errors from compounding.

The stability loss term is defined as

$$\mathcal{L}_{\text{stability}} = \frac{1}{N_t} \sum_{t=1}^{N_t} \sum_{v \in V} (\mathcal{A}(u^{(t)} + \epsilon) - u^{(t+1)})^2, \quad (28)$$

where $u^{(t)}$ is the ground truth state (water depth and volume) at time t . The term ϵ represents a small perturbation introduced to approximate the shifted input distribution encountered during inference. In other words, the perturbed input is meant to provide a possible realization of $\mathcal{A}(u^{(t-1)})$. This means that instead of relying on perfect ground truth inputs at every step, the model conditions its predictions on previously predicted states. This mimics the exact scenario the model faces at inference, where it must predict the next state from its previous prediction. This loss encourages the model to correct for small perturbations in its own predictions, making it more robust to autoregressive errors.

In practice, the pushforward trick is implemented by modifying the training procedure to introduce perturbed inputs. Instead of training only on ground truth data, the model also receives predictions from previous timesteps, which include accumulated errors. The model is unrolled for two steps, meaning it first predicts $\hat{u}^{(t)}$, and then uses that prediction to generate $\hat{u}^{(t+1)}$, instead of always conditioning on true states. To ensure robust generalization, gradients are computed and backpropagated only from the last step. This prevents the model from memorizing true values at early steps, making it more resilient to errors that arise during inference. This process forces the model to make stable predictions even when its inputs are slightly off to better handle real-world forecasting conditions.

The total loss function used for training the model combines the regular one-step prediction loss with the stability loss:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{one-step}} + \mathcal{L}_{\text{stability}}, \quad (29)$$

where $\mathcal{L}_{\text{one-step}}$ ensures that predictions match ground truth values as closely as possible, while $\mathcal{L}_{\text{stability}}$ ensures that small deviations in predictions do not destabilize future forecasts. By unrolling the model for two steps but backpropagating only on the final step, both computational efficiency and model stability are achieved, dampening error propagation.

4.5 | KAN for interpretability

The KAN framework enhances interpretability by decomposing each input feature into univariate functions using



Fourier series expansions (J. Xu et al., 2024). In HydroGraphNet, KAN layers are used in the encoder to process node features before they are passed into the GN layers. Specifically, for each node v , let $x_{v,p}$ represent the p th feature, where p indexes specific attributes such as slope, elevation, or Manning's roughness coefficient. The KAN representation for each feature $x_{v,p}$ is given by

$$\text{KAN}(x_{v,p}) = \sum_{k=1}^K a_{p,k} \cos(kx_{v,p}) + b_{p,k} \sin(kx_{v,p}), \quad (30)$$

where k indexes the harmonic terms in the Fourier expansion, up to a maximum K . The coefficients $a_{p,k}$ and $b_{p,k}$ are learnable parameters associated with the p th feature's k th harmonic term.

By mapping each feature through univariate Fourier expansions, the KAN's transformations can be visualized to reveal how the model *thinks* about specific ranges of each environmental variable. For instance, if a slope transformation peaks at specific values, it indicates that terrains with such slopes exert a stronger effect on predicted flood depths. These univariate mappings feed into subsequent GN layers, shaping every water-depth and volume forecast. Importantly, this structured decomposition not only enhances interpretability but also implicitly contributes to physically plausible predictions by capturing intrinsic, physics-based relationships, such as the natural association between lower elevations or concave terrains and higher flood susceptibility. By examining the learned transformations, one can see which ranges of elevation or roughness most influence flood propagation, and why a particular node (mesh cell) is predicted to have a certain water depth. Consequently, the KAN makes it possible to disentangle the contributions of each input variable, granting a clear view of how terrain characteristics and hydraulic properties jointly drive the model's predictions.

5 | RESULTS AND DISCUSSION

This section presents the results from experiments conducted with the proposed HydroGraphNet model for flood forecasting in dynamic civil infrastructure systems. The model's performance is evaluated in predicting water depth and volume across multiple time steps, comparing its accuracy, mass conservation adherence, and interpretability against a baseline GNN model. Mathematical formulations for the performance metrics and model evaluation techniques are included for clarity.

5.1 | Experimental setup

The proposed model utilizes an autoregressive encoder–processor–decoder GNN architecture to capture the spatiotemporal dynamics of flood forecasting while ensuring adherence to physical laws. Table 2 summarizes the key components and hyperparameters of the architecture.

To assess the performance of the proposed framework, its results are compared against two models: (1) a *baseline GNN model*, which does not incorporate KAN layers in the node encoder or the physics-based loss (instead, a standard MLP is used for the node encoder). The push-forward trick is integrated into the baseline GNN to isolate and assess the advantages of incorporating the physics-based loss; and (2) a *UNet model* adapted for flood prediction. The UNet model maps the irregular computational mesh onto a structured grid, utilizing a CNN-based encoder–decoder architecture (Ronneberger et al., 2015). This comparison highlights the impact of the physics-informed loss, the interpretability offered by KAN, and the advantages of graph-based unstructured representations over grid-based approaches. The evaluation is conducted on a case study of the White River, focusing on a section north of Muncie, Indiana. A two-dimensional hydrodynamic model in hydrologic engineering center-river analysis system (HEC-RAS) is used, configured for rain-on-grid modeling, where rainfall is applied uniformly over mesh cells to simulate precipitation. The base hydrographs and hyetographs are derived from historical flow records of the Iowa River (U.S. Geological Survey, 2024) and weather observing stations (National Centers for Environmental Information, NOAA, 2024). To create diverse input conditions, these base hydrographs and hyetographs are scaled with random factors drawn from a uniform distribution, generating synthetic boundary conditions and rainfall events. Figure 3a shows the base hydrographs, Figure 3b presents the hyetographs, and Figure 3c illustrates the 4787 HEC-RAS computation points distributed across a mesh with an 80-m resolution.

A total of 525 simulations are generated by varying the scaled hydrographs and rainfall patterns. The dataset is divided into 85% for training, 10% for validation, and 5% for testing. Simulations are conducted with a spatial resolution of 80 m (Δx) and a computational time interval of 20 s (Δt) in HEC-RAS to satisfy the CFL condition and ensure stable flood wave propagation. However, this high-resolution approach can be computationally prohibitive for machine learning purposes, so the data are downsampled to a 20-min resolution for model training and evaluation to balance accuracy and feasibility. Both the proposed and benchmark models perform autoregressive rollouts, where predictions from previous steps serve

**TABLE 2** Details of HydroGraphNet architecture and hyperparameters.

Component	Configuration
Encoder ($\mathcal{E}$)	Node: KAN, harmonics: 5, hidden dimension: 64, layers: 1 Edge: MLP, hidden dimension: 64, layers: 3
Processor ($\mathcal{P}$)	Five GN blocks: Node MLP: hidden dimension: 64, layers: 3 Edge: MLP: hidden dimension: 64, layers: 3
Decoder ($\mathcal{D}$)	MLP: hidden dimension: 64, layers: 3
Learning rate	10^{-4} , exponential decay (0.9999979 per epoch)
Training duration	300 epochs
Computational platform	NVIDIA RTX 4090 GPU
Temporal resolution	20 mins
Batch size	One

Abbreviations: KAN, Kolmogorov–Arnold network; MLP, multi-layer perceptron.

as inputs for subsequent steps, as depicted in Figure 4, enabling continuous flood propagation forecasting.

Hyperparameter tuning is conducted using Optuna, leveraging Bayesian optimization (Snoek et al., 2012) to minimize validation errors. Models are trained using the Adam optimizer (Kingma & Ba, 2014) over 300 epochs. Training was performed on an NVIDIA RTX 4090 GPU, with batch sizes of 1. This configuration ensures the models can process the full spatial domain while maintaining scalability for larger datasets.

To assess model performance, four primary metrics are used:

1. **Root mean square error (RMSE):** RMSE quantifies predictive accuracy by measuring the deviation between predicted and observed water depths across nodes. It is calculated as

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{h}_i - h_i)^2}, \quad (31)$$

where $\hat{h}_i$ and h_i are the predicted and observed water depths at node i and N is the total number of nodes. A lower RMSE indicates higher predictive accuracy.

2. **Nash–Sutcliffe efficiency (NSE) coefficient:** The NSE coefficient evaluates the model's predictive skill by comparing predictions to the mean of observed values. It is defined as

$$\text{NSE} = 1 - \frac{\sum_{i=1}^N (\hat{h}_i - h_i)^2}{\sum_{i=1}^N (h_i - \bar{h})^2}, \quad (32)$$

where $\bar{h}$ is the mean observed water depth. An NSE closer to 1 indicates better model performance, while values below 0 suggest the model performs worse than the average observed value.

3. **Mass balance error:** Mass balance error measures the conservation of mass across the domain by comparing predicted water volumes at successive time steps. The error, $\mathcal{E}_{\text{mass}}$, at time t is given by

$$\mathcal{E}_{\text{mass}}(t) = \left(\sum_{i=1}^N v_i^{(t)} - \sum_{i=1}^N v_i^{(t-1)} - \Delta t \cdot \left(\frac{s^{(t-1)} + s^{(t)}}{2} - \sum_{i=1}^N \frac{i_i^{(t-1)} + i_i^{(t)}}{2} \cdot A_i \right) \right)^2, \quad (33)$$

where $v_i^{(t)}$ is the predicted volume at node i at time t , $s^{(t)}$ represents the inflow, and Δt is the time interval between steps. A lower mass balance error reflects better adherence to mass conservation principles, which is crucial for physically consistent flood forecasting.

4. **Critical success index (CSI):** CSI evaluates the model's ability to correctly predict significant flood events by assessing the overlap between predicted and observed exceedances of a threshold that is chosen based on practical flood risk management considerations. In our study, two thresholds are used: 0.3 and 0.05 m. The 0.3-m threshold represents a critical depth for issuing flood warnings in many operational contexts, while the 0.05-m threshold is used to capture lower magnitude flooding events that are still of interest for early detection. The CSI is defined as

$$\text{CSI} = \frac{H_{\text{thr}}}{H_{\text{thr}} + M_{\text{thr}} + \text{FA}_{\text{thr}}}, \quad (34)$$

where H_{thr} represents the number of hits (instances where the water depth at a node exceeds the threshold in both the GNN model and the hydrodynamic model), M_{thr} represents the number of misses (instances where the water depth exceeds the threshold only in the hydrodynamic model), and FA_{thr} represents the num-

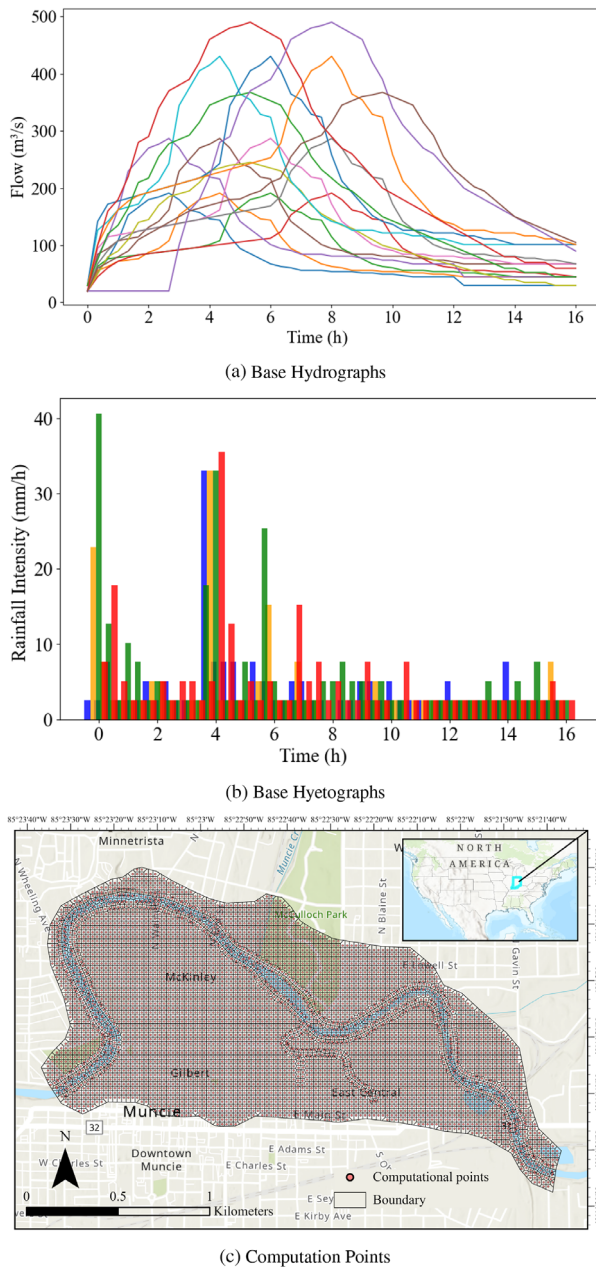


FIGURE 3 (a) Base hydrographs showing flow rates over time. (b) Base hyetographs showing rainfall intensity over time. (c) Computation points used in the HEC-RAS simulation (4787 points).

ber of false alarms (instances where the water depth exceeds the threshold only in the GNN model). A higher CSI value indicates better performance in predicting significant flood events above the specified thresholds.

5.2 | Results and analysis

Table 3 provides a comparative assessment of the three models—HydroGraphNet, baseline GNN, and UNet—across key performance metrics and trainable parameters. The HydroGraphNet consistently outperforms the base-

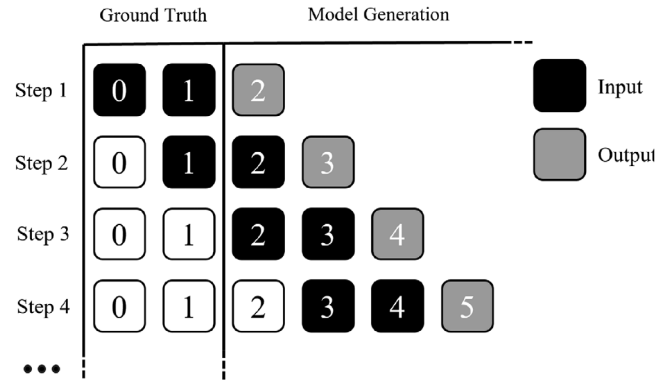


FIGURE 4 Example rollout using a GNN-based surrogate model, where the model predicts one timestep per iteration using an input sequence of two timesteps. For the initial rollout step, the GNN model relies entirely on ground-truth timesteps from the original simulation. In subsequent rollout steps, the input increasingly includes predictions generated by the model,

line GNN and UNet, achieving the best results in terms of RMSE and NSE, which reflect its superior accuracy and reliability, while its parameter count remains comparable to that of the baseline GNN and is much lower than the UNet. Additionally, the HydroGraphNet demonstrates the highest CSI for both lower and higher flood depth thresholds, highlighting its enhanced ability to detect moderate and severe flood events. In comparison, the baseline GNN struggles, particularly at the higher threshold, indicating limitations in capturing extreme flood conditions. The UNet model performs relatively well at the lower threshold but is overall less effective, with higher RMSE and lower NSE than the HydroGraphNet despite its significantly larger number of trainable parameters. Together, these results validate the HydroGraphNet as a more accurate and robust model for predicting and detecting flood events across various intensity levels. Figure 5 compares the HydroGraphNet and baseline GNN models at three time steps: 2, 7, and 15 h into the flood event. The spatial distribution of predicted water depths is juxtaposed against ground truth data, and absolute error maps highlight discrepancies between predictions and observations. Both models operate in an autoregressive manner, relying on their own previous predictions as inputs at each time step, and both incorporate the pushforward trick to mitigate error accumulation. However, despite this, the baseline GNN exhibits larger errors in the river region, particularly at the 15-h mark, where distinct vertical error patterns emerge. These high-error regions result from the accumulation of prediction errors over successive time steps, primarily due to the baseline model's lack of explicit physics constraints. Without enforcing physical laws such as mass conservation, the baseline GNN gradually deviates from physically consistent flood dynamics, leading to unrealistic predictions that pushforward alone is unable to correct.

TABLE 3 Comparison of model performance metrics (mean $\pm$ standard deviation) averaged over all test scenarios and time steps and the number of trainable parameters.

Model	RMSE	NSE	CSI		# Parameters
			thr = 0.05	thr = 0.3	
HydroGraphNet	0.18 ± 0.09	0.89 ± 0.13	0.93 ± 0.03	0.87 ± 0.07	165,058
Baseline GNN	0.55 ± 0.30	0.44 ± 0.94	0.84 ± 0.06	0.55 ± 0.18	160,130
UNet model	0.27 ± 0.18	0.83 ± 0.03	0.87 ± 0.04	0.76 ± 0.12	471,114

Abbreviations: CSI, critical success index; GNN, graph neural network; RMSE, root mean square error; NSE, Nash–Sutcliffe efficiency; thr, threshold.

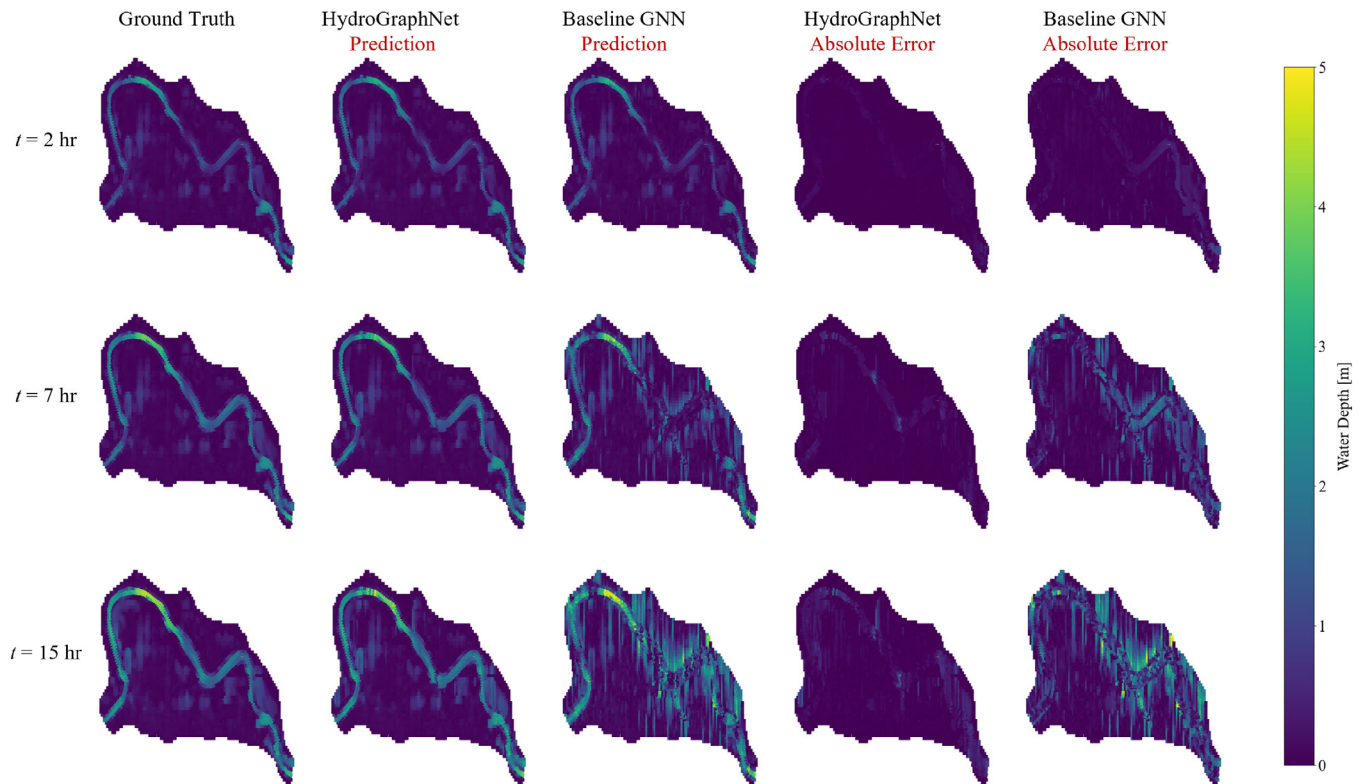


FIGURE 5 Comparison of water depth predictions at 2, 7, and 15 h. Each row shows ground truth, HydroGraphNet predictions, baseline GNN predictions, and corresponding absolute errors. Both models rely on their own previous predictions in an autoregressive manner, but the baseline GNN exhibits increased error in the river region, particularly at 15 h, due to the accumulation of prediction errors over time.

This effect is especially pronounced in high-variability regions like the river channel, where small deviations cascade into significant discrepancies. In contrast, HydroGraphNet maintains high accuracy even at the 15-h mark, closely matching the ground truth water depths. This superior performance stems from its physics-informed design that leverages a physics-based loss to rigorously enforce mass conservation and a KAN module to extract physically relevant patterns, collectively yielding predictions that align with the underlying physical principles.

Figure 6a shows that the RMSE, computed over time and averaged across all test hydrograph scenarios, is consistently lower for the HydroGraphNet compared to the baseline model. This improvement is particularly notable at later stages, where cumulative errors in the baseline

become increasingly significant. This outcome suggests that integrating physics constraints in the GNN effectively minimizes error accumulation over time. At the final time step, the average RMSE for the HydroGraphNet is 0.28 m, compared to 1.16 m for the baseline model.

To further assess the models' capability to predict significant flood events, the CSI serves as an additional performance metric. Figure 6b presents the mean CSI over time for both models at the 0.3-m threshold. The HydroGraphNet consistently achieves higher CSI values compared to the baseline model at the 0.3-m threshold, indicating superior performance in detecting significant flood events. At the final time step, the HydroGraphNet achieves an average CSI of 0.81, whereas the baseline model reaches a CSI of 0.43.

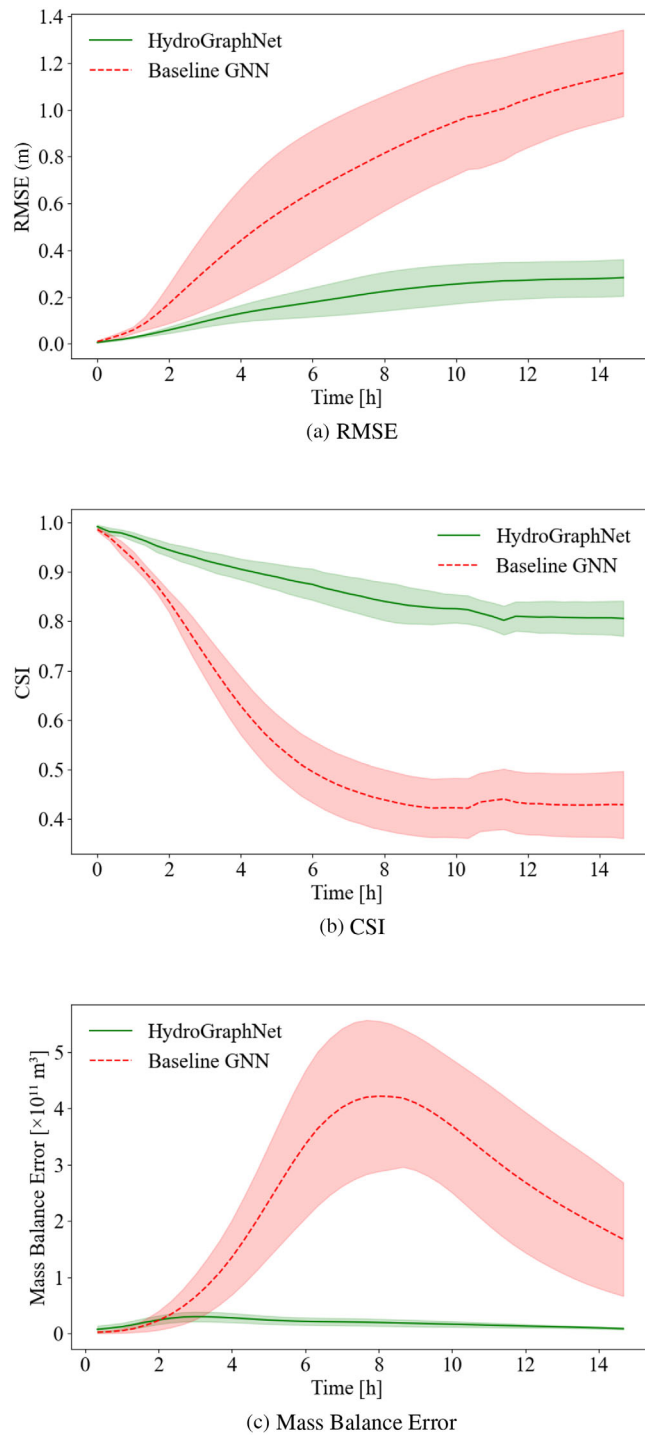


FIGURE 6 Comparison of performance metrics over time for physics-informed and baseline GNN models: (a) RMSE, (b) CSI at threshold 0.3 m, and (c) mass balance error. Shaded regions denote the standard deviation of each metric computed on the test dataset.

5.2.1 | Mass balance error

The mass balance error, $\mathcal{E}_{\text{mass}}$, is computed at each time step to assess the conservation of mass, with a value of zero indicating perfect adherence. Figure 6c

shows that the baseline GNN exhibits substantial mass balance violations, with errors reaching up to 4×10^{11} , highlighting significant deviations from conservation principles. In contrast, HydroGraphNet maintains a near-zero mass balance error across time steps, which demonstrates its effectiveness in enforcing physical consistency. This strict adherence to mass conservation enhances the reliability of HydroGraphNet's flood predictions.

5.2.2 | Flood arrival time

Beyond predictive accuracy and mass balance, each model's capability is evaluated to forecast flood arrival time, defined as the moment when flood water depth reaches 0.3 m. This metric is essential for assessing the onset of flooding, which is critical for effective early warning and response in vulnerable areas.

Figure 7 compares flood arrival times predicted by the baseline GNN, HydroGraphNet, and the ground truth obtained from hydrodynamic simulations. The baseline model inaccurately predicts flood extent and timing due to error accumulation in its autoregressive rollout. Specifically, it predicts flooding over a larger portion of the domain, including many areas that remain unflooded according to the ground truth. On the other hand, HydroGraphNet shows closer alignment with the ground truth, accurately capturing the spatial extent of flooding and correctly identifying regions with early flood arrival, while greatly reducing false-positive predictions in unaffected regions. This result further emphasizes the advantage of incorporating physics into GNN architectures, through both the physics-informed loss and the KAN module that captures physically relevant patterns, to enhance accuracy in flood arrival time predictions.

Figure 8 further supports these findings by showing the ratio of the flooded area over time. The baseline GNN exhibits significant deviations from the ground truth, overestimating the flood spread, particularly in later time steps. HydroGraphNet, on the other hand, closely follows the ground truth, accurately capturing the temporal evolution of flooded areas. This consistency underscores its effectiveness in modeling flood spread dynamics with improved physical realism.

Accurate flood arrival time predictions are crucial for applications where both timing and depth play a vital role, such as issuing flood warnings, optimizing emergency evacuation routes, and managing real-time flood mitigation efforts. HydroGraphNet's ability to provide timely and accurate forecasts makes it particularly well-suited for these critical applications.

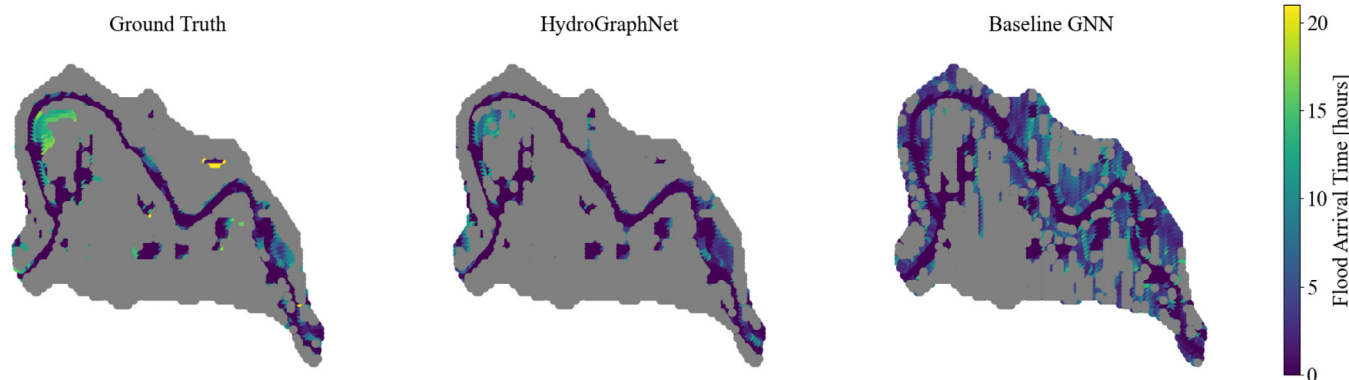


FIGURE 7 Spatial comparison of flood arrival time between models and ground truth.

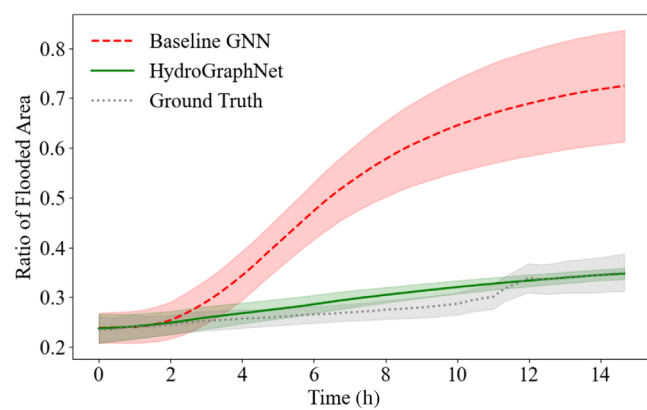


FIGURE 8 Ratio of the flooded area over time, comparing baseline GNN, HydroGraphNet, and ground truth. Shaded areas represent the standard deviation of the metric computed on the test dataset.

5.2.3 | Interpretability with KAN transformations

The KAN framework in the HydroGraphNet enhances interpretability by mapping complex environmental features into univariate transformations and revealing their distinct roles in flood dynamics. Figure 9 illustrates the KAN transformations for key features—elevation, curvature, Manning's roughness coefficient, and slope—offering insights into how the model interprets and weights these variables in predicting flood behavior.

The transformation for elevation (Figure 9a) demonstrates a decreasing trend, with higher transformed values assigned to lower elevations. This suggests that the model prioritizes low-lying areas, which are naturally more prone to water accumulation and flooding while deemphasizing higher elevations that contribute to drainage. By focusing on lower elevations, the model aligns with hydrological principles, capturing the vulnerability of these areas to flood events.

For curvature (Figure 9b), the transformation assigns greater importance to concave regions, which are characterized by negative curvature values, while convex regions are downweighted. Concave areas are more likely to retain water, which makes them critical in flood modeling, whereas convex areas tend to facilitate runoff. This transformation indicates that the model effectively distinguishes between regions that contribute to water pooling and those that promote drainage.

The transformation for Manning's roughness coefficient (Figure 9c) presents a U-shaped response, highlighting both very smooth and highly rough surfaces. This dual emphasis likely reflects the contrasting roles of surface roughness in water flow; smooth surfaces allow for faster flow, whereas rough surfaces increase resistance, slowing down water and potentially promoting local accumulation. By assigning significance to both extremes of surface roughness, the model adapts to the variable impacts of terrain texture on water movement.

The slope transformation (Figure 9d) follows a bell-shaped curve, with a peak around moderate slope values (approximately 25° – 30°). This does not imply that moderate slopes universally promote stagnation or impede drainage. Rather, in our study area (the White River near Muncie), these slopes frequently correspond to riverbanks and transitional zones between the main channel and the floodplain. During flood events, these boundary regions play a critical role in determining how water spreads into the floodplain or recedes back into the channel. By contrast, very flat areas are more prone to prolonged inundation, while steep slopes primarily act as drainage pathways with less water retention. Consequently, the model's learned transformation emphasizes moderate slopes because they mediate the exchange of water between the river and the floodplain, thereby influencing flood dynamics in these transitional zones.

Together, these transformations underscore the model's capacity to adapt to critical terrain features by weighting

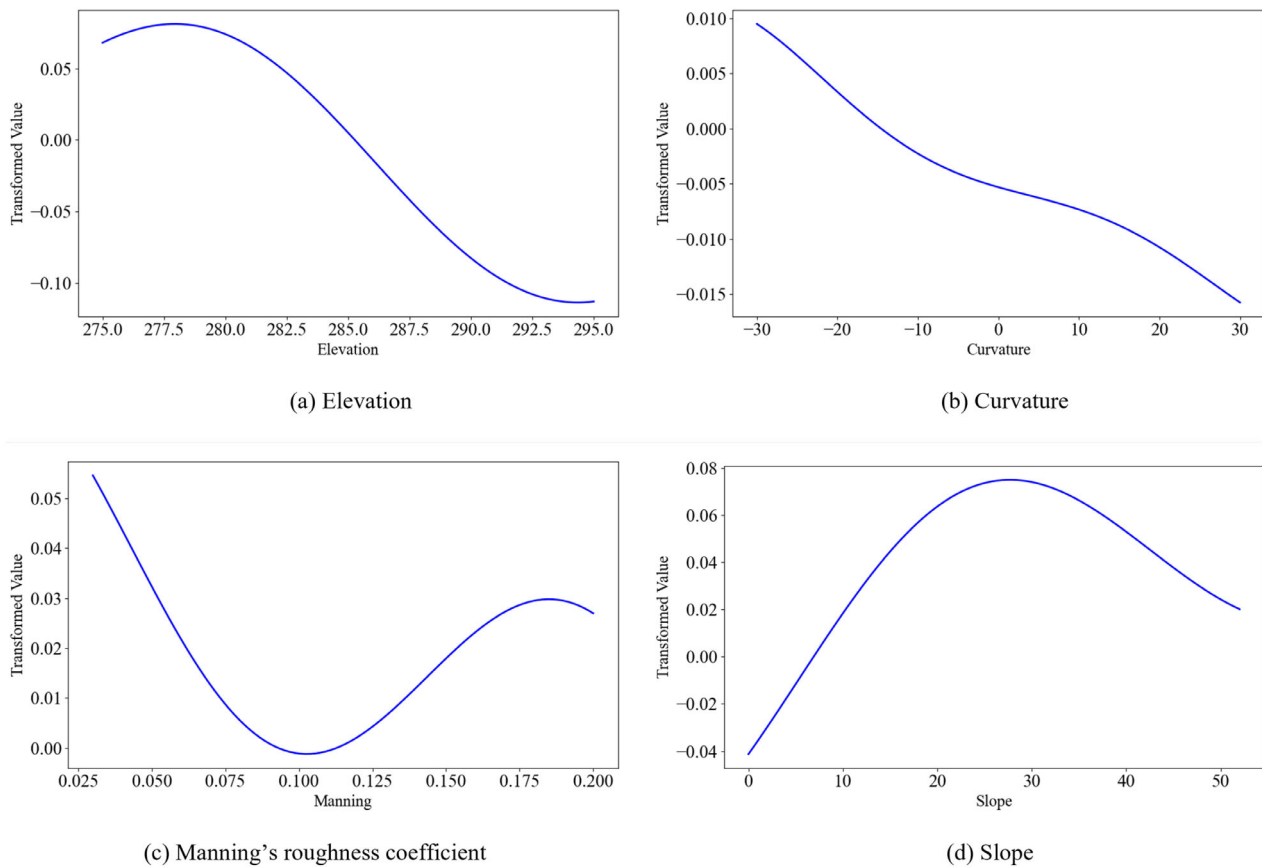


FIGURE 9 KAN transformations for selected environmental features. (a) Elevation, (b) curvature, (c) Manning's roughness coefficient, and (d) slope.

TABLE 4 Comparison of HydroGraphNet performance with and without KAN.

Model	RMSE	NSE	CSI	
			thr = 0.05	thr = 0.3
Without KAN	0.27 ± 0.15	0.80 ± 0.16	0.88 ± 0.04	0.77 ± 0.11
With KAN	0.18 ± 0.09	0.89 ± 0.13	0.93 ± 0.03	0.87 ± 0.07

Abbreviations: CSI, critical success index; KAN, Kolmogorov–Arnold Network; RMSE, root mean square error; NSE, Nash–Sutcliffe efficiency; thr, threshold.

each variable according to its hydrological relevance. This targeted weighting of elevation, curvature, surface roughness, and slope enables the model to generate more accurate and physically consistent flood predictions, as it assigns importance based on the flood-contributing characteristics of each environmental factor.

Beyond interpretability, the integration of KAN in HydroGraphNet also enhances model performance by better capturing the physically meaningful relationships between environmental variables and flood dynamics, which in turn implicitly reinforces physically plausible predictions. To empirically assess its impact, we compare the model's accuracy with and without KAN, as shown in Table 4. The results show consistent improvements across

all metrics, which underscores KAN's role in both enhancing complex environmental interaction representation and upholding physical consistency.

5.2.4 | Computational efficiency

While demonstrating enhanced predictive accuracy and improved physical consistency compared to the baseline GNN, the HydroGraphNet maintains comparable computational efficiency. Inference times for the HydroGraphNet and baseline GNN are 0.2 and 0.12 s, respectively, on an NVIDIA RTX 4090 GPU for a 15-h forecast. This marginal computational overhead (0.08 s) is offset by the HydroGraphNet's superior predictive performance. Both GNN variants exhibit a substantial computational advantage over traditional physics-based simulations: the equivalent HEC-RAS simulation requires approximately 130 s under comparable hardware conditions. The HydroGraphNet achieves a speedup of over three orders of magnitude relative to conventional numerical methods while retaining the baseline GNN's real-time inference capabilities. This enables rapid scenario analysis and time-sensitive decision-making.



6 | CONCLUSION

This study presents HydroGraphNet, a novel physics-informed GNN framework for flood forecasting, specifically designed to address the limitations of traditional hydrodynamic models and purely data-driven approaches. By embedding mass conservation principles into the loss function, the framework ensures physical consistency in predictions while maintaining computational efficiency. The model leverages an autoregressive encoder–processor–decoder architecture to capture complex spatiotemporal flood dynamics, enhanced by the KAN for interpretability. The KAN decomposition not only reveals the influence of environmental features such as elevation, curvature, slope, and Manning's roughness coefficient on flood behavior but also reinforces physically consistent transformations that align with hydrological principles. Extensive experiments on the White River case study demonstrate that the HydroGraphNet achieves superior accuracy and stability compared to a baseline GNN. Key performance improvements include a 67% reduction in RMSE, a near-zero mass balance error, and a 58% higher CSI for significant flood events. The model accurately forecasts flood arrival times and dynamically evolving flood patterns, closely aligning with high-fidelity hydrodynamic simulations.

The proposed framework addresses critical challenges in flood forecasting by combining data efficiency, scalability, and physical interpretability, making it particularly suitable for real-world applications. Future work could focus on integrating multifidelity modeling (Taghizadeh, Nabian, et al., 2024) into the framework to leverage low-fidelity data to reduce the volume of high-fidelity training data required while maintaining predictive accuracy. Another key future direction is to enhance scalability. Our modular framework can be integrated with distributed and multiscale GNN strategies, such as domain decomposition with halo exchange (Nabian et al., 2024) and distributed graph construction (Barwey et al., 2024), to efficiently handle higher resolution, large-scale flood models.

Additionally, a potential limitation of the current approach is the limited generalizability of the model once trained to different case studies. As part of future work, the training dataset will be extended and transfer learning (Y. Xu et al., 2023) along with domain adaptation techniques (Zhou & Pan, 2022) will be incorporated. These enhancements are expected to improve the model's ability to adapt to varying river basin profiles and diverse hydrological and geographical conditions.

DATA AVAILABILITY STATEMENT

The implementation of HydroGraphNet, along with the dataset generation code and processed dataset, is publicly available through NVIDIA PhysicsNeMo at https://github.com/NVIDIA/physicsnemo/tree/main/examples/weather/flood_modeling/hydrographnet.

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